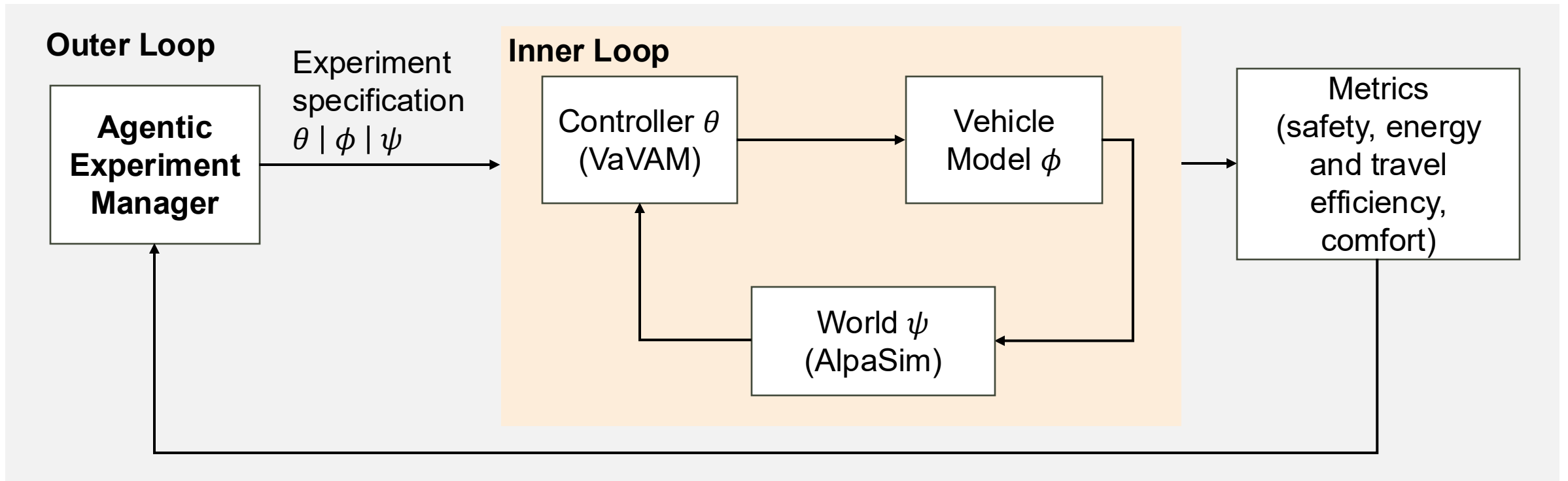
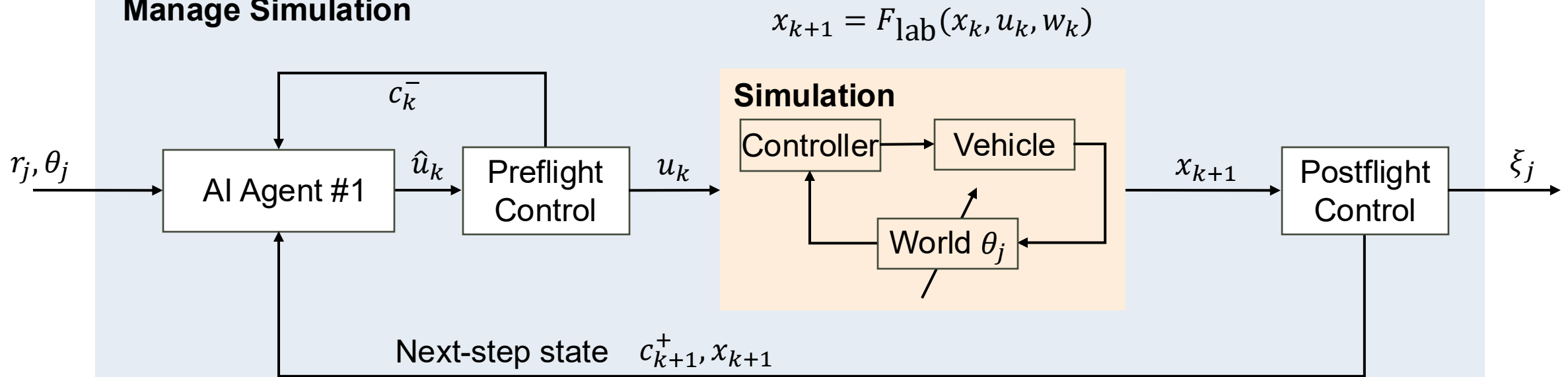




Scientific question: How can a hierarchical feedback-control architecture simultaneously ensure reliable AI-agent execution and sample-efficient experiment adaptation?

- **Concept:** Closed-loop autonomous experimentation: specify $\rightarrow$ simulate $\rightarrow$ evaluate $\rightarrow$ adapt.
- **Two agent roles (see next slides):**
 - **Agent #1:** execute and recover individual simulations.
 - **Agent #2:** manage experiment-level iteration and select the next test.
- **Inner loop:** vehicle–controller–world interaction for one experiment.
- **Outer loop:** adapt experiment parameters using measured outcomes.

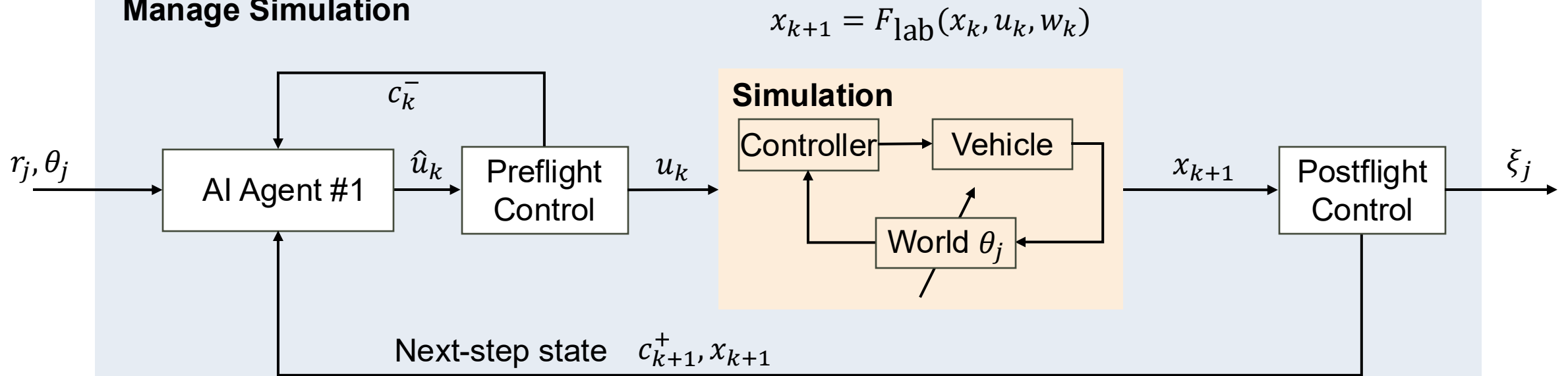
AI Agent Loop #1: Manage Simulation



Symbol	Meaning
x_k	Experiment state (discrete)
ξ_k	Physical states (continuous)
r	Experiment request
$\hat{u}_k$	Agent-proposed action
u_k	Approved executed action
c_k^-	Preflight feedback
c_{k+1}^+	Postflight feedback
w_k	Execution uncertainty

Step	Update	Meaning
1	$\hat{u}_k \sim \pi_A(x_k, r, c_k^+, c_k^-)$	Agent proposes from state and feedback
2	$c_k^- = K^-(x_k, r, \hat{u}_k)$	Preflight control scores the proposal
3	$u_k = \hat{u}_k$ if $c_k^- = \text{APPROVE}$	Loop repeats until APPROVE
4	$x_{k+1} = F_{\text{lab}}(x_k, u_k, w_k)$	Simulation advances the state
5	$c_{k+1}^+ = K^+(x_k, u_k, x_{k+1}, r)$	Postflight feedback for the next step

AI Agent Loop #1: Manage Simulation

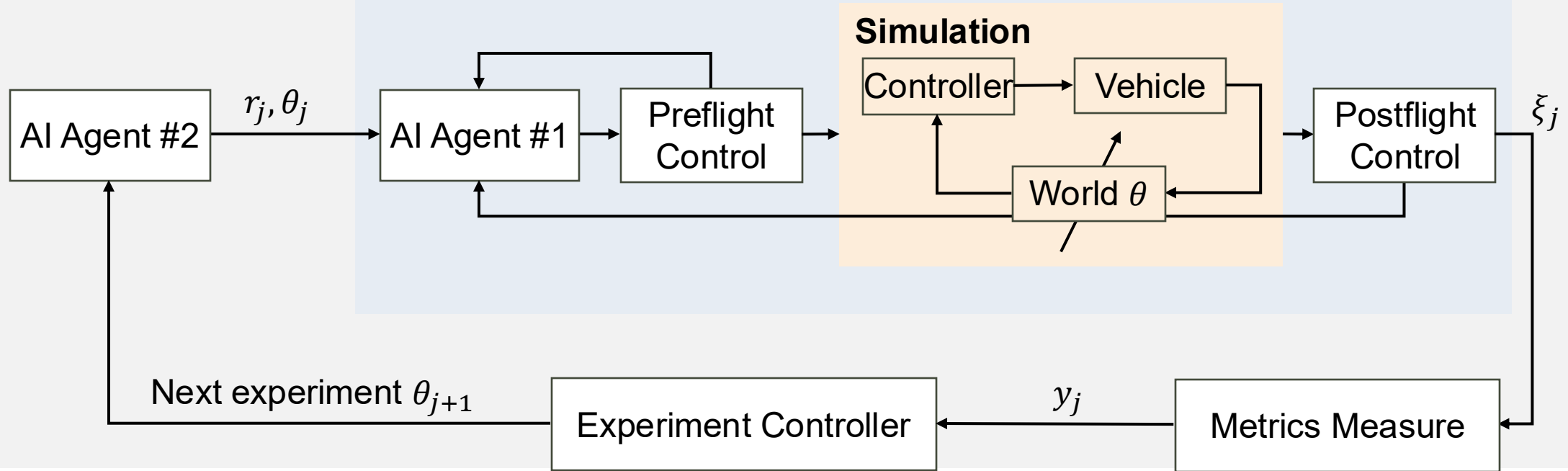


Every variable ranges over a finite set; the elements below are proposed, not yet determined.

Variable	Finite domain	Proposed elements (illustrative, not fixed)
$x_k \in \mathcal{X}$	Experiment state (discrete)	IDLE, CONFIGURED, RUNNING, PAUSED, COMPLETED, FAULTED
$\xi_j \in \mathcal{E}$	Physical states (continuous)	Ego and world trajectories, recorded signals
$r \in \mathcal{R}$	Experiment request	TRACK_SPEED, CHANGE_LANE, FOLLOW_VEHICLE, EMERGENCY_STOP
$\mathcal{S}$	Agent skills / tools	CONFIGURE, LAUNCH, INSPECT, STOP, RESTART, VALIDATE, RECOVER
$u_k = (s_k, \theta_k)$	Executed action (skill, params)	$s_k \in \mathcal{S}$, $\theta_k \in \Theta(s_k)$
$c_k^- \in \mathcal{C}^-$	Preflight feedback	APPROVE, REVISE_PARAMETER, CHANGE_SKILL, REQUEST_INFO, ABORT
$c_{k+1}^+ \in \mathcal{C}^+$	Postflight feedback	SUCCESS, PARTIAL, NO_EFFECT, CONSTRAINT_HIT, FAULT
$w_k \in \mathcal{W}$	Execution uncertainty	NOMINAL, SENSOR_NOISE, ACTUATOR_LAG, DISTURBANCE, DROPOUT

AI Agent #2: Automate Experiment

AI Agent Loop #1: Manage Simulation



Symbol	Meaning
j	Experiment iteration index
$r_j \in \mathcal{R}$	Experiment request to Agent #1
$\theta_j \in \mathcal{\Theta}$	Experiment parameters for run j
$\xi_j \in \mathcal{E}$	Physical states: ego and world trajectories
$y_j \in \mathcal{Y}$	Experiment performance metrics
$h: \mathcal{E} \rightarrow \mathcal{Y}$	Metrics map
$C_{\text{exp}}: \mathcal{\Theta} \times \mathcal{Y} \rightarrow \mathcal{\Theta}$	Experiment controller

Step	Update	Meaning
1	$(r_j, \theta_j) \rightarrow \text{Loop \#1}$	Agent #2 sets experiment + parameters
2	$\xi_j = \mathcal{E}(r_j, \theta_j)$	Loop #1 runs it, returns a result
3	$y_j = h(\xi_j)$	Metrics block scores the output
4	$\theta_{j+1} = C_{\text{exp}}(\theta_j, y_j)$	Controller picks next parameters
5	$(r_j, \theta_{j+1}) \rightarrow \text{Loop \#1}$	Only parameters advance; repeat